iRAS

Recirculating Aquaculture System (RAS)

Engineering Design Technical Manual

Version 1.9

Revision 1.9 — carbonate-equilibrium water-quality model (pH, CO_2 speciation, un-ionised NH_3, buffer intensity β, calcite saturation Ω); the strippable CO_2 load now includes nitrification; species- and salinity-aware CO_2 limits and alkalinity targets.

Process design · Mass balance · Unit sizing

Heat balance · Multi-module N+1 · P&ID generation

Equipment schedule · Investment & financial evaluation · Feasibility report

Polarlys Innovation AS · Cheng Sun

iras.yis.no · github.com/chengsunmail/iras-en · polarlysinnovation@gmail.com

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PART I

Fundamentals

Chapter 1 Introduction to Recirculating Aquaculture

1.1 What is a RAS?

A Recirculating Aquaculture System (RAS) is an intensive aquaculture technology in which the culture water is physically, chemically and biologically treated and then reused. Unlike traditional flow-through culture, a RAS requires only a small daily make-up of new water (typically 5–15% per day); the remaining 85–95% is treated and recirculated.

The core idea is to continuously remove the metabolic wastes produced by the fish — total ammonia nitrogen (TAN), suspended solids, dissolved organics and carbon dioxide — while replenishing dissolved oxygen, so that water quality is held within the optimal range for healthy growth inside a closed or semi-closed loop.

A typical RAS comprises the following core components: culture tanks (fish or shrimp tanks), a solid–liquid separation unit (rotary drum filter, RDF), a biofilter (MBBR moving-bed biofilter), a disinfection unit (UV), an oxygenation unit (LOX system or aeration), and a degassing unit (CO_2 stripping column).

1.2 RAS versus traditional aquaculture

Dimension	Pond culture	Flow-through	RAS
Water use	Evaporation make-up	Large intake + discharge	Only 5–15%/day make-up
Stocking density	0.5–5 kg/m^3	10–30 kg/m^3	30–100 kg/m^3
Water-quality control	Natural purification	Dilution by flow	Precise process control
Footprint	Large	Medium	Small (1/10–1/50 of a pond)
Environmental impact	Diffuse pollution	Discharge pollution	Zero or minimal discharge
Siting	Near a water source	Near a water source	Almost unrestricted
Capital cost	Low	Medium	High
Operating cost	Low	Low	Medium–high (power / LOX)

Dimension	Pond culture	Flow-through	RAS
Product consistency	Variable	Fairly stable	Most stable
Biosecurity	Low	Low	High (closed)

1.3 Advantages and challenges

Advantages

- **Water saving:** 85–95% reuse; only 0.5–1 m^3 of new water per kg of fish, versus 5–20 m^3 for traditional ponds.
- **Land saving:** high density (30–100 kg/m^3); footprint only 1/10–1/50 of a pond.
- **Controllable:** temperature, DO, pH and TAN are controlled year-round, independent of season and weather.
- **Environmentally sound:** wastewater is treated centrally, solids can be reused as fertiliser, and there is no diffuse pollution.
- **Biosecure:** the closed system excludes external pathogens and reduces medication.
- **Flexible siting:** independent of a natural water body — can be built near cities, in deserts or inland.

Challenges

- **High capital cost:** equipment on the order of hundreds of EUR per m^3; total project investment on the order of thousands of EUR per m^3.
- **Operating cost:** electricity and liquid oxygen are the main costs, typically 15–30% of total cost.
- **Technical threshold:** requires multidisciplinary knowledge; operators need professional training.
- **System fragility:** a power or equipment failure can kill fish within hours — backup power is essential.
- **Slow biofilter start-up:** the nitrifier community takes 4–8 weeks to establish.

1.4 Industry status

- **Norway / Scotland:** the world's largest RAS salmon industry; single plants exceed 10,000 t/yr (Atlantic Sapphire's Miami plant is planned at 100,000 t/yr).
- **Denmark:** a major exporter of RAS technology and equipment (AKVA group, Billund Aquaculture).
- **Spain (Stolt Sea Farm):** >50% of European turbot production; industrial densities of 70–100 kg/m^2 of floor area.

- **China:** rapid growth since 2010, mainly turbot, grouper, largemouth bass and whiteleg shrimp.
- **Japan:** a leader in eel RAS.

1.5 Typical application scenarios

Scenario	Typical species	Density (kg/m^3)	Temp (°C)	Notes
Cold-water plant	Salmon, trout	50–80	12–15	Freshwater fry + seawater grow-out, long cycle
Warm-water plant	Largemouth bass, mandarin fish	40–60	24–28	Strong domestic demand, low FCR
Marine plant	Turbot, grouper	30–70	16–27	Needs seawater / brackish water
Eel plant	Japanese / European eel	40–60	26–28	Powder-feed fed
Shrimp plant	Whiteleg shrimp	2–15	28–31	Low density but high feeding rate

Chapter 2 RAS Process Flow

2.1 Standard RAS process flow

In a standard RAS the water recirculates in the following order:

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Fish tank → Drum filter (RDF) → [Skimmer] → Biofilter (BF) → UV → CO₂ stripper →
Oxygenation → Fish tank Side-streams: [Denitrification] / [AOP]
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Square brackets [ ] denote optional units, installed depending on species and water-quality requirements. In iRAS this flow is modelled precisely as a **hybrid of a series mainline and parallel side-streams**: each unit's removal efficiency is computed independently and the concentration decreases step by step along the loop.

2.2 Function of each treatment unit

Unit	Target	Principle	Typical efficiency
Drum filter (RDF)	TSS (suspended solids)	60–90 µm screen, physical interception	60–90%
Protein skimmer	DOM + fine TSS	Foam fractionation, optional ozone	DOM 40–70%
MBBR biofilter	TAN (ammonia)	Nitrifiers oxidise	60–90%

Unit	Target	Principle	Typical efficiency
		NH_4 to NO_3	
Denitrification reactor	NO_3 (nitrate)	Anoxic reduction of NO_3 to N_2	50–80%
UV	Bacteria / virus / parasites	UV damages DNA	99–99.9%
AOP (advanced oxidation)	Geosmin / 2-MIB (off-flavour)	Ozone + UV synergy	80–95%
Oxygenation cone / aeration	DO (dissolved oxygen)	LOX dissolution or air aeration	110–165% saturation
CO_2 stripper	CO_2 (carbon dioxide)	Counter-current air stripping	60–85% (G:L = 3–7)

Target oxygen saturation is recommended by species group: salmonids 165%, warm-water marine fish 140%, warm-water freshwater carnivores 110–125%, tilapia / shrimp 100–115%.

2.3 Freshwater versus seawater systems

- **Lower DO saturation:** seawater (30‰, 15°C) saturates at about 8.4 mg/L, ~17% lower than freshwater.
- **Reduced nitrification:** salinity inhibits nitrifiers; correction factor $f = \max(1 - 0.01 \times S, 0.3)$.
- **Ozone → bromate risk:** Br^- in seawater forms carcinogenic BrO_3^- under ozone.
- **Corrosion:** 316L stainless or HDPE/FRP required, raising equipment cost 30–50%.
- **Better skimming:** seawater's higher surface tension yields more stable foam, so skimming is 20–30% more effective than in freshwater.

Chapter 3 Water-Quality Parameters and Control Standards

3.1 Key water-quality parameters

3.1.1 Total ammonia nitrogen (TAN)

TAN is the main nitrogenous waste of protein metabolism, excreted through the gills. In water it exists as ionised NH_4^+ and molecular NH_3, whose ratio depends on pH and temperature. NH_3 is strongly toxic; the higher the pH and temperature, the larger the NH_3 fraction. RAS TAN is typically held at 0.5–2.0 mg/L, with NH_3 below 0.02–0.05 mg/L.

3.1.2 Nitrite (NO_2-N)

An intermediate of nitrification, second only to NH_3 in toxicity. NO_2 should be kept below 0.5 mg/L. Persistently high values indicate an incomplete nitrifier community (insufficient Nitrobacter), usually during start-up.

3.1.3 Nitrate (NO_3-N)

The end-product of nitrification — low toxicity but it accumulates in a closed loop. Sustained high NO_3 (>100–200 mg/L) suppresses growth. Denitrification or water exchange are the control measures.

3.1.4 Dissolved oxygen (DO)

Both fish and nitrifiers need ample DO. Tank DO is typically required to be >6 mg/L (cold-water fish) or >5 mg/L (warm-water fish). Pure-oxygen systems can raise DO to 110–165% saturation (super-saturation) to support high density. iRAS uses a dual threshold: max(satRatio × DO_sat, DO_abs) — e.g. salmonids 0.75 + 6.0 mg/L, tilapia 0.60 + 4.0 mg/L. DO saturation is computed precisely with the Benson & Krause (1984) formula (see Chapter 7).

3.1.5 Carbon dioxide (CO_2)

Fish respiration produces CO_2; in a closed system it accumulates, lowering pH and stressing fish. RAS CO_2 is typically held below 15 mg/L (seawater) or 20 mg/L (freshwater) by counter-current air stripping. Salmonids are more CO_2-sensitive than this split implies: the Norwegian smolt standard is <15 mg/L (FOR 2004; Bergheim et al. 2009), with growth effects reported from ~12 mg/L and nephrocalcinosis near 16, so iRAS holds salmonids to 15 regardless of (freshwater) salinity. CO_2 and pH are coupled through the carbonate system and solved together per stage (§4.5).

3.1.6 Suspended solids (TSS)

Includes uneaten feed, faeces and sloughed biofilm. High TSS clogs gills, consumes oxygen and harbours harmful bacteria. RAS TSS is typically held below 15 mg/L.

3.1.7 Dissolved organic matter (DOM)

Dissolved proteins, fatty acids and humic substances. DOM is not intercepted by the RDF nor effectively removed by the biofilter, so it accumulates, yellowing the water, causing foaming and producing off-flavour. Protein skimming and AOP are the main DOM-removal routes.

3.2 Per-species tolerance ranges

Parameter	Salmon	Bass / mandarin	Tilapia	Shrimp	Grouper	Turbot	Eel
TAN (mg/L)	<1.0	<1.5	<2.0	<1.5	<1.0	<1.0	<1.5
NO_2-N	<0.3	<0.5	<1.0	<1.0	<0.5	<0.5	<0.5
NO_3-N	<80	<100	<200	<100	<80	<80	<100
DO	>6	>5	>4	>5	>5	>6	>5
CO_2	<15	<20	<20	<15	<15	<15	<20
TSS	<10	<15	<25	<15	<15	<10	<15
pH	6.5–7.5	6.8–7.8	6.5–8.5	7.5–8.5	7.5–8.2	7.5–8.2	7.0–8.0

Parameter	Salmon	Bass / mandarin	Tilapia	Shrimp	Grouper	Turbot	Eel
Temp (°C)	10–16	22–28	25–32	28–32	24–30	14–18	24–28
Salinity (‰)	0→30	0	0–15	5–35	25–35	25–35	0–5

3.3 Consequences of water-quality deterioration

The core of RAS water management is **stability** — fluctuation within the safe range is far more dangerous than the absolute value:

- **TAN/NH_3 exceedance:** acute toxicity and chronic immunosuppression — the most common cause of death in RAS.
- **NO_2 exceedance:** “brown-blood disease” (haemoglobin oxidised to methaemoglobin), causing hypoxia.
- **DO deficiency:** fish gasp at the surface; below 3 mg/L, mass mortality within hours.
- **Excess CO_2:** blood acidosis; impairs calcification (moulting difficulty in shrimp).
- **Excess TSS:** gill clogging and bacterial growth.
- **pH swing:** nitrification consumes alkalinity and drives pH down; below 6.5 nitrifier activity collapses.

Chapter 4 Nitrogen and Carbon Cycles

Understanding the nitrogen cycle is the basis of process design. All nitrogen originates in feed protein, passes through a series of biochemical transformations and ultimately leaves the system as N_2 gas.

4.1 From feed protein to nitrogen

After feeding, dietary protein is digested and assimilated:

- About 25–35% of feed nitrogen is assimilated into fish biomass (growth).
- About 50–60% is excreted as TAN through the gills.
- About 10–15% is excreted as faeces (particulate organic nitrogen, intercepted by the RDF).

The iRAS TAN production formula:

TAN (kg/d) = daily feed × protein fraction × tanCoef

where tanCoef defaults to **0.092** (Timmons & Ebeling 2010). Marine carnivores are slightly higher (turbot 0.092, grouper 0.110, mandarin 0.100).

4.2 Nitrification

Nitrification is the most critical biological process in a RAS, carried out by two groups of autotrophic bacteria under aerobic conditions:

$NH_4 + 1.5\ O_2 \rightarrow NO_2 + 2H + H_2O$ (Nitrosomonas) $NO_2 + 0.5\ O_2 \rightarrow NO_3$ (Nitrobacter) Overall: $NH_4 + 2\ O_2 \rightarrow NO_3 + 2H + H_2O$

From an engineering standpoint, three stoichiometric relations matter:

- **Oxygen demand:** 4.57 g O_2 per g TAN-N oxidised.
- **Alkalinity consumption:** 7.14 g $CaCO_3$ equivalent per g TAN-N.
- **Acid production:** H^+ generation lowers pH — the fundamental reason RAS pH falls.
- **Nitrate production (steady-state mass balance):** ~1 g NO_3-N per g TAN-N (≈ 1:1).

Steady-state nitrate production Under steady state the NO_3 production rate ≈ TAN production rate × 1.0 (≈ 100% nitrification), **not** × η_bio. Physically, at steady state the fish-produced TAN equals the TAN accumulated through biofilter nitrification plus the TAN exported by water exchange; since exchange typically exports <1% of the fish TAN, "all TAN becomes NO_3" holds within engineering accuracy. In iRAS the biofilter nitrification oxygen demand is supplied by a separate coarse-bubble aeration and does not draw on the tank's main oxygenation budget. This **allocate-O_2-by-location** principle avoids the double-counting of oxygen common in traditional calculations.

4.3 Denitrification

Under anoxic conditions, heterotrophs use an external carbon source (methanol) to reduce NO_3 to N_2 gas — the only process that truly removes nitrogen from the loop:

$6\ NO_3 + 5\ CH_3OH \rightarrow 3\ N_2\uparrow + 5\ CO_2 + 7\ H_2O + 6\ OH$

- **Methanol consumption:** 2.47 g methanol per g NO_3-N reduced (including heterotroph growth).
- **Alkalinity recovery:** 2.8 g $CaCO_3$ per g NO_3-N (iRAS engineering default). The stoichiometric limit is 3.57 — exactly half of nitrification's 7.14 — but part of the recovered alkalinity is offset by cell synthesis and incomplete reduction, so measured values fall at 2.5–3.0 (median 2.8). Overridable per stage (denitriAlkRecovery).

In iRAS denitrification is a side-stream: 5–15% of the recirculating flow is diverted into an anoxic reactor and returned to the mainline. The influent NO_3 concentration is taken at the **biofilter outlet** (P&ID convention). A well-designed denitrification branch lowers NO_3 and recovers alkalinity, cutting $NaHCO_3$ dosing by 30–50%.

4.4 Alkalinity balance

$$\text{Net alkalinity demand} = \text{TAN\_nitrified} \times 7.14 - NO_3\text{\_denitrified} \times 2.8 - \text{exchange} \times \text{source\_alk}$$

If the net demand is positive, an alkalinity source must be dosed to hold pH:

$$NaHCO_3 \text{ dose (kg/d)} = \text{alkalinity deficit} / 0.595$$

where 0.595 = 50/84 is the $NaHCO_3$-to-$CaCO_3$ conversion factor. The exchange compensation uses V_total (the total system volume), not V_tank.

4.5 Carbonate equilibrium: pH, CO_2 speciation and un-ionised ammonia

Tank pH is not a free parameter — it is fixed by the carbonate system. Total alkalinity (TA, held by $NaHCO_3$ dosing) and the dissolved CO_2 produced by respiration and nitrification together determine pH, the CO_2 / HCO_3^- / CO_3^{2-} split, the un-ionised ammonia fraction and the calcite saturation state. iRAS solves this equilibrium per stage with a dedicated carbonate module.

Governing relations. From TA and aqueous CO_2 the module solves the proton balance for $[H^+]$ and back-calculates the full speciation. Dissociation constants follow Millero (2010) over salinity 0–50 and temperature 0–40 °C (carbonic acid K_1/K_2, plus the borate, water and bisulfate systems used for the pH-scale conversions). The internal calculation is on the seawater pH scale; the reported pH is converted to the **free (concentration) scale**, which equals the seawater scale at zero salinity and matches a freshwater probe reading.

Un-ionised ammonia. The toxic species NH_3 is computed from the ammonia dissociation constant of Clegg & Whitfield (1995); the NH_3 fraction of total ammonia (TAN) rises sharply with pH and temperature. The per-stage NH_3-N limit is species-specific (Table 3.2) and is halved for fry / early-life stages.

Buffer intensity and calcite saturation. The module returns the Van Slyke buffer intensity β (mg/L $CaCO_3$ per pH unit), which quantifies how strongly alkalinity damps a pH excursion after feeding, and — in seawater — the calcite saturation Ω, relevant to shell-forming species and to scaling.

Two control modes. Each stage runs in one of two modes: *hold alkalinity* (the default — the operator sets a target alkalinity and pH emerges) or *hold pH* (the operator sets a target pH and the module inverts the equilibrium to report the alkalinity required). The default alkalinity target is salinity-aware: 150 mg/L $CaCO_3$ in freshwater, 200 in seawater. Higher alkalinity in marine systems is justified because it lowers CO_2, stabilises pH and aids nitrification (Letelier-Gordo et al. 2024).

Validation. The module was validated against $PyCO_2SYS$ 1.8.3 across salinity 0–35 and temperature 12–28 °C: forward pH agrees to within 0.001 unit, the inverse alkalinity solve to 0.08 µmol/kg, and the NH_3 fraction and calcite Ω to within rounding.

These are daily-average, steady-state (chronic-exposure) values. The short post-feeding transient is damped by the buffer β and bounded by the peak-sized CO_2 stripper, so — unlike dissolved oxygen, which is sized on the instantaneous post-feeding peak — CO_2, pH and NH_3 are reported on a steady-state basis.

PART II

Engineering Design Methods

Chapter 5 Farm-Scale Design

The first step in RAS engineering design is to back-calculate the system scale from the target annual yield. This determines the sizing basis for every downstream water-treatment unit.

5.1 Little's Law and steady-state modelling

iRAS models a continuous, staggered-batch culture: the farm stocks in staggered batches so that market-size fish leave every day. At steady state the system simultaneously holds fish of different sizes across each stage. Little's Law (the fundamental theorem of queueing) applies:

$$L = \lambda \times W$$

where L = average number in the system (individuals), λ = arrival rate (individuals/yr), W = residence time (yr). The key is that Little's Law applies to **individual counts**, not biomass. A salmon grows from 0.1 g to 5000 g — a 50,000× change in mass; using mass would grossly overestimate the standing population of the early stages.

The steady-state standing count includes a mortality correction:

$$N\_steady = N\_inflow \times (months / 12) \times (1 - mortality/2)$$

5.2 Count conservation and loss-rate recursion

Counts in a multi-stage RAS are recursed from the last stage backwards, amplified at each stage by the loss rate:

Annual harvest count = annual yield (kg) / market individual weight (kg) This-stage inlet count = next-stage inlet count / (1 − this-stage loss rate)

The loss rate covers mortality, culling and escape, and is usually highest in the fry stage (10–30%) and lowest at grow-out (2–5%).

5.3 Three feed definitions

iRAS strictly separates three time-scales, each used for a different purpose:

① Daily peak feed (feedMax) — the load-calculation basis

feedMax = biomass × feedRate (kg/d)

biomass is the steady-state standing stock and feedRate is the maximum daily feeding rate at the end of the stage. All daily pollutant production (TAN/TSS/DOM/CO_2) is computed from feedMax (without peakFactor).

② Hourly peak (xxxPeakHourly) — the equipment-sizing basis

xxxPeakHourly = xxxDaily / 24 × peakFactor

peakFactor (default **1.5**, Timmons & Ebeling 2010) is the ratio of the hourly peak metabolic rate to the daily average. Fast-kinetics equipment (tank oxygenation, biofilter blowers) is sized on xxxPeakHourly × safety.

③ Cycle daily-average (feedAvg) — the annualised-OPEX basis

feedAvg = yearGrowth × FCR / 365 (kg/d)

where yearGrowth is the annual flesh gain of the stage and FCR is the stage feed-conversion ratio.

Peak-to-average check Peak-to-average ratio = feedRate / (feedAvg/biomass). If this ratio is below 1.1, equipment may be undersized and iRAS issues a warning.

5.4 Equipment classification by kinetics

Class	Equipment	Sizing basis	Physical reason
Slow kinetics	Biofilter, denitrification, CO_2 stripper, drum filter, skimmer, UV	Daily load × safety	Biofilm metabolism has inertia; hours of HRT buffer the peak
Fast kinetics	Tank oxygenation, biofilter blower	Hourly peak × safety	Oxygen has no storage; must meet the instantaneous peak demand

Chapter 6 Pollutant Load Calculation

Once feed is determined, the daily production of each pollutant follows. iRAS strictly distinguishes the daily, hourly-peak and daily-average fields.

6.1 TAN production

TAN_Daily = feedMax × protein × tanCoef TAN_PeakHourly = TAN_Daily / 24 × peakFactor TAN_DailyAvg = feedAvg × protein × tanCoef

tanCoef defaults to 0.092 (Timmons & Ebeling 2010); marine carnivores are slightly higher (0.10–0.11).

6.2 TSS and DOM production

TSS_Daily = feedMax × 0.30 DOM_Daily = feedMax × 0.10

6.3 Oxygen demand — three components by location

O_2 source	Formula	Supplied by	Notes
① Fish respiration	feedMax × o2FishFactor	Tank main oxygenation	Species-specific; Q_{10}-scaled from the species' metabolic reference temperature (same reference as the heat model)
② Nitrifier demand	TAN_Daily × 4.57	Biofilter coarse aeration	$NH_4 \rightarrow NO_3$ stoichiometry
③ Heterotroph demand	feedMax × o2DOMfactor	Biofilter coarse aeration	Default 0.10 (Boyd 2018)

Key design principle Tank main oxygenation (①) and biofilter aeration (②+③) are two independent systems and do not overlap. Merging all three and supplying them entirely from the tank oxygenation would either starve the biofilter or over-design the tank oxygenation.

6.4 CO_2 production

Dissolved CO_2 in the loop has two sources — fish respiration and nitrification:

$$CO_2\_Daily = o2FishDaily \times 1.375 + TAN\_Daily \times 6.286$$

The first term is respiratory: 1.375 = 44/32 is the CO_2:O_2 molar-mass ratio at RQ ≈ 1. The second term is nitrification: oxidising ammonium acidifies the water ($NH_4^+ + 2\ HCO_3^- + 2\ O_2 \rightarrow NO_3^- + 2\ CO_2 + 3\ H_2O$), so the two bicarbonate ions consumed per mole of TAN-N are converted to CO_2 — 2 × 44.01 / 14.01 = 6.286 g CO_2 per g TAN-N. (Earlier manual versions omitted this term, which is material at high feed loads.)

Denitrification adjustment. Where a denitrification branch is active, the recovered alkalinity reconverts some free CO_2 back to bicarbonate while the co-oxidised methanol releases CO_2; the net effect is a small reduction of −0.147 g CO_2 per g $CaCO_3$ of alkalinity recovered, which iRAS subtracts from the strippable load.

Load basis versus metabolic carbon. The figure above is the *strippable* CO_2 load that sizes the stripper and sets pH — for salmon ≈ 0.5 g CO_2 per g feed (respiration ≈ 0.3 + nitrification ≈ 0.2). It is deliberately higher than, and not directly comparable to, the *metabolic* total-inorganic-carbon production (≈ 0.345 g/g) sometimes quoted in the literature: the metabolic figure counts only the fish, whereas the design figure is the gas that must actually be stripped to hold pH.

Chapter 7 Exact Recirculation Equations

This is the core calculation engine of iRAS. Traditional RAS design often estimates the steady-state concentration with the additive approximation $C = \Delta / \Sigma\eta$, but when the system contains several series and side-stream units the additive form produces significant error. iRAS uses exact multiplicative recirculation equations so that mass conservation holds strictly.

7.1 Mass-balance derivation

Consider a steady-state loop: water leaves the tank, passes n treatment units, and returns. Each unit has its own pass-through fraction $(1 - \eta)$. Let Δ = the concentration increment the tank adds per cycle (mg/L), C_in = treated return concentration (tank inlet, lowest), C_out = tank outlet concentration (highest), and R = total pass-through = the product of all units' pass-through fractions. At steady state:

$C\_out = C\_in + \Delta$ (tank adds the increment) $C\_in = C\_out \times R$ (after full treatment, back to C_in)

Solving simultaneously:

$C\_out = \Delta / (1 - R)$ (tank outlet — the sizing basis) $C\_in = \Delta \times R / (1 - R)$ (return water — the lowest concentration)

7.2 Total pass-through R

$R = \Pi(1 - \eta) \times (1 - \text{dilution})$

The smaller R (the higher the total removal), the lower C_out and the better the water quality.

7.3 Series versus side-stream modelling

Series pass-through (RDF / BF / CO_2 stripper)

Full flow passes through; note it is a product:

$\text{series pass-through} = (1-\eta\_rdf)(1-\eta\_bio) = 0.25 \times 0.30 = 0.075$

i.e. 92.5% total removal. The additive form 0.75 + 0.70 = 1.45 > 100% is physically impossible.

Side-stream pass-through (skimmer / AOP / denitrification)

Taking the skimmer as an example, with side-stream flow ratio α = 20% and single-pass η = 70%:

$\eta\_sys = \alpha \times \eta\_single = 0.20 \times 0.70 = 14\%$

Physically, 80% of the water bypasses while 20% is treated at 70% removal → 14% overall after mixing.

7.4 Closure verification

An elegant property of the exact equations: after the concentration decreases step by step along the loop, the final return concentration is exactly C_in. iRAS verifies this closure condition internally after every calculation (error = 0).

7.5 R path per parameter

Parameter	Treatment path	R expression
TAN	BF nitrification + exchange	$(1-\eta_{bio})(1-d)$
TSS	RDF + skimmer(side) + BF + exchange	$(1-\eta_{rdf})(1-\eta_{skim})(1-\eta_{bioTSS})(1-d)$
DOM	skimmer(side) + AOP(side) + exchange	$(1-\eta_{skim\_sys})(1-\eta_{aopDOM\_sys})(1-d)$
NO_3	denitrification(side) + exchange	$(1-\eta_{denitri\_sys})(1-d)$
CO_2	stripper + exchange	$(1-\eta_{co2})(1-d)$
Geosmin	AOP(side) + exchange	$(1-\eta_{aop\_sys})(1-d)$

7.6 Dilution sensitivity

The influence of the dilution term differs greatly between parameters. When a parameter has a dedicated treatment unit (TAN, TSS, CO_2), η dominates and V_total affects the result by <1%. When a parameter relies on water exchange alone (NO_3 with no denitrification, Geosmin with no AOP, DOM with no skimmer), dilution dominates: with $\eta = 0$, R = 1 − dilution and C_out = Δ/dilution, so doubling the dilution halves the concentration. This is precisely why correct system-volume modelling and a consistent exchange-rate definition matter.

Chapter 8 System-Volume Modelling

8.1 Why model the process water volume?

Traditional RAS design often simplifies the whole system to the "tank water volume", ignoring:

- Biofilter (MBBR) water, typically 10–20% of V_tank
- Drum-filter sump ≈ Q × short HRT
- CO_2 stripper water ≈ degasArea × 1.2 m
- Oxygenation-cone water ≈ Q × 30 s HRT
- AOP / denitrification / skimmer side-stream water
- Collection sump
- Pipework + buffer ≈ V_tank × 5%

These process volumes typically sum to 30–60% of V_tank and cannot be ignored.

8.2 V_process component formulas

Process unit	Formula	Basis
V_RDF	Q × rdfHRT_min / 60	Drum-filter sump HRT, default 0.5 min (compact 0.3; traditional 1.5–2)
V_BF (incl. media water)	bioVolume / fillRatio	fillRatio default 0.5 (Rusten 2006) → V_BF = 2 × bioVolume
V_skim	skimmerVolume	Skimmer water volume (already a water volume)
V_UV	5 m^3 (fixed)	Flow-through UV, negligible internal water
V_AOP	aopVolume	AOP reactor water volume
V_denitri	denitriVolume	Denitrification reactor
V_co2	degasArea × co2VolFactor	Stripper water, default factor 0.7 (sump 0.5 m + media 0.2 m)
V_aerator	Q × 0.5 / 60	Oxygenation cone HRT ≈ 30 s
V_sump	clamp(Q/30, 20, 200)	Collection sump, engineering convention
V_pipes	V_tank × 5%	Pipework + buffer, engineering experience

Deriving V_BF When fillRatio = 50% (typical MBBR), 100 m^3 of net media corresponds to a 200 m^3 total tank volume (half media, half interstitial water). The total tank volume (media + water) is the effective water that participates in steady-state dilution, so V_BF = bioVolume / fillRatio.

8.3 V_total and steady-state concentration

V_total = V_tank + V_process

Effect on the steady-state concentration C_out = $\Delta/(1-R)$: the per-cycle increment Δ is independent of V (V cancels), but the dilution term inside R depends on V_total. Therefore V_total's influence depends on the treatment route — with a dedicated unit (TAN/TSS/CO_2) η dominates and the influence is <1%; with exchange-only dilution (NO_3 without denitrification) the concentration scales as V_tank/V_total.

8.4 Exchange-rate definition

The user-entered “exchangeDaily” (e.g. 5%/d) is defined against the **total system volume** (V_total), consistent with the Timmons textbook and large industrial projects. Actual exchange volume = V_total × exchangeDaily.

8.5 Worked example: salmon grow-out, 1000 t/yr

Process unit	Volume (m^3)
Tank water V_tank	3,385
BF (incl. water)	570
CO_2 stripper	350
RDF sump	338
Collection sump	200
Pipework + buffer	169
Denitrification	129
Oxygenation cone	85
Skimmer	74
AOP reactor	17
UV	5
V_process total	**1,938** (57% of V_tank)
V_total = V_tank + V_process	**5,323**

PART III

Treatment-Unit Design

Chapter 9 Solid–Liquid Separation (Rotary Drum Filter, RDF)

9.1 Principle

The rotary drum filter (RDF) is the first line of defence in a RAS, physically intercepting suspended solids on a 60–90 µm stainless screen. Water flows from the inside of the drum outward; solids are retained on the screen. When the pressure differential across the screen reaches a set point, the RDF’s built-in backwash mechanism starts automatically. The backwash pump is **not** listed as a separate item, because all commercial RDFs (Hydrotech, Faivre, etc.) ship with an integral backwash pump, and the backwash water is largely returned to the loop via a sludge settling tank (Sharrer 2010: 95–98% clarified-water return).

9.2 Design parameters

Parameter	Typical value	Notes
Screen aperture	60–90 µm	Smaller aperture = higher removal but clogs sooner

Parameter	Typical value	Notes
TSS removal	60–90%	iRAS default 75%
Throughput	recirc flow × safetyRDF	Default safety 1.10
RDF sump HRT	0.1–5 min	iRAS default 0.5 min (median)

9.3 Design formulas

RDF throughput = recirc flow × safetyRDF (m^3/h) RDF sump V = flow × rdfHRT_min / 60 (m^3)

9.4 System discharge

The true liquid crossing the system boundary is determined by the water-exchange rate, not by the backwash:

System discharge (m^3/d) = V_total × exchangeDaily System-discharge TSS (kg/d) = load.tssDaily (mass conservation, incl. sludge solids) System-discharge TAN_eff (mg/L) = C_tank (dissolved steady-state concentration) System-discharge NO_3_eff (mg/L) = C_tank (dissolved steady-state concentration)

TSS basis The iRAS "system-discharge TSS" is the **settling-tank inlet** concentration (sludge-laden, mass-conservation basis), not the discharge-pipe concentration. For a 1000 t salmon case TSS ≈ 3257 mg/L, a reasonable settling-tank inlet value (typical RAS range 1000–5000 mg/L). It must not be compared directly with pipe-outlet effluent standards. Settling-tank design and supernatant treatment are handled separately by the wastewater discipline; the iRAS boundary stops at the settling-tank inlet.

9.5 Faults and maintenance

Symptom	Possible cause	Action
Effluent TSS high	Screen damaged / aperture enlarged	Replace screen (life 2–3 yr)
Frequent backwash	High inlet TSS / clogged screen	Clean screen; check feeding
Backwash water clear	Nozzle clogged / low pressure	Clean nozzles; check the integral backwash mechanism
Drum not turning	Motor fault / broken chain	Check motor and drive

Chapter 10 Biofiltration (MBBR Biofilter)

10.1 Nitrification principle

The MBBR is the most critical treatment unit in a RAS. HDPE moving-bed media (specific surface area 500–800 m^2/m^3) are kept in suspension by aeration; nitrifiers attached to their surface oxidise the toxic TAN to low-toxicity NO_3. Nitrification is a slow autotrophic process; the community takes 4–8 weeks to establish.

10.2 Volumetric TAN removal rate (VTR)

VTR_{20} = 0.3–0.8 kg TAN/(m^3·d) @ 20°C (iRAS default 0.5) $VTR(T) = VTR_{20} \times \theta^{(T-20)}$ where $\theta = 1.10$

Temp (°C)	Correction θ^(T−20)	VTR kg/(m^3·d)	Notes
10	0.386	0.193	Cold-water fry
12	0.467	0.234	Salmon fry
15	0.621	0.310	Salmon grow-out
16	0.683	0.341	Turbot full cycle
20	1.000	0.500	Baseline
25	1.611	0.805	Bass / mandarin fish
28	2.144	1.072	Tilapia
30	2.594	1.297	Shrimp

Salinity correction

$f\_sal = \max(1 - 0.01 \times S, 0.3)$

At 30‰ salinity, nitrification efficiency falls to 70%.

10.3 Media-volume calculation

BF media volume = TAN_Daily × safetyBio / [$VTR_{20} \times \theta^{(T-20)} \times f\_sal$]

10.4 Biofilter coarse-bubble aeration

BF O_2 = TAN × 4.57 + feedMax × o2DOMfactor Airflow = O_2_Peak × safetyBFblow / (1.2 × 0.23 × SOTE) SOTE = 8% Blower power = airflow(m^3/s) × 50000 / (0.60 × 1000) (50 kPa, $\eta = 0.60$)

o2DOMfactor defaults to 0.10 (Boyd 2018). The biofilter blower carries the P&ID tag **K-302**.

10.5 Start-up management

Biofilter start-up takes 4–8 weeks; mismanagement causes TAN/NO_2 spikes and mass mortality. Steps:

- **Week 1:** empty tank, dose nitrifier inoculum, add NH_4Cl to TAN 2–4 mg/L, hold 25–30°C.
- **Weeks 2–3:** monitor TAN/NO_2; a falling TAN shows Nitrosomonas is working.
- **Weeks 3–5:** NO_2 falls and NO_3 appears, showing Nitrobacter has established.
- **Weeks 5–8:** TAN < 0.5 and NO_2 < 0.3 sustained for 3 days marks maturity.
- **Week 8+:** ramp load gradually, no more than 20% per week.

Cautions: hold pH 7.0–7.5 (nitrification consumes alkalinity; below 6.5 activity collapses); keep biofilter DO > 2 mg/L; never use antibiotics (they kill nitrifiers); never stock to full load at once.

Chapter 11 Protein Skimmer

11.1 Principle

A protein skimmer removes DOM and fine TSS (<30 µm) by foam fractionation. Fine air or ozone bubbles are injected into a contact column; surface-active organics adsorb onto the bubble surfaces and overflow with the foam.

11.2 Air versus ozone

Parameter	Air mode	Ozone mode
DOM single-pass removal	30–50% (default 40%)	60–80% (default 70%)
TSS single-pass removal	~40%	~40%
Extra effect	None	Disinfection + decolouring + oxidation
Cost	Low	High (ozone generator)
Seawater risk	None	Bromate BrO_3^- (control needed if salinity >5‰)

11.3 Design calculation

η_sys = flow ratio × single-pass efficiency = 0.20 × 0.70 = 14% Skimmer V = side-stream flow (m^3/h) × HRT (min) / 60 × safetySkimmer Ozone demand = feedMax × 0.010 g O_3 / g feed

HRT defaults to 2 min. DOM adsorption onto bubbles is a seconds-scale reaction; HRT mainly ensures stable foam formation, so a longer HRT brings no benefit. The skimmer carries tag **S-701**.

Chapter 12 Denitrification Reactor

12.1 Principle

Under anoxic conditions (DO < 0.5 mg/L) heterotrophs use methanol to reduce NO_3 to N_2 gas — the only process that truly removes nitrogen from a RAS:

$$6\ NO_3 + 5\ CH_3OH \rightarrow 3\ N_2\uparrow + 5\ CO_2 + 7\ H_2O + 6\ OH$$

12.2 Design calculation

Side-stream flow = recirc flow × flow ratio (default 10%) η_sys = flow ratio × single-pass efficiency = 10% × 60% = 6% VDR(T) = VDR_{20}(0.8) × 1.08^(T−20) Reactor volume = NO_3 removed × safetyDenitri / VDR(T) Methanol = NO_3 removed × 2.47 (kg/d) Alkalinity recovery = NO_3 removed × 2.8 (kg $CaCO_3$/d)

The reactor carries tag **R-801**. Methanol is a flammable hazardous chemical and must be stored separately; the denitrification reactor starts up faster than the biofilter (1–2 weeks).

Chapter 13 UV Disinfection

13.1 Principle

UV-C (254 nm) penetrates microbial cell walls and breaks DNA thymine dimers, destroying the ability to replicate. Advantages: no chemical residue, no change to water chemistry, fast kill.

13.2 Chick–Watson model

$$\text{inactivation} = 1 - \exp(-k \times D)$$

Pathogen	k (cm^2/mJ)	Inactivation @ D=40	Recommended dose
Bacteria (E. coli, Vibrio)	0.10–0.20	99.7%	30–40 mJ/cm^2
Fish viruses (IPNV, IHNV)	0.05–0.10	86–98%	60–100 mJ/cm^2
Protozoa (Ich cysts)	0.15–0.25	99.7–99.99%	30–40 mJ/cm^2
Saprolegnia spores	~0.30	99.999%	20–30 mJ/cm^2

13.3 UV power formula

$$\text{UV power (W)} = \text{throughput} \times \text{UV dose} \times 0.85 \times \text{safetyUV}$$

safetyUV is the rated-power margin for lamp ageing, not a dose increase. The UV unit carries tag **D-401**. Maintenance: log lamp hours (life 8000–12000 h); wipe the quartz sleeves monthly with dilute acid; keep TSS < 10 mg/L upstream.

Chapter 14 Advanced Oxidation (AOP)

14.1 Principle

Ozone + UV synergistically generate hydroxyl radicals (·OH) that oxidise and break down geosmin and 2-MIB (off-flavour compounds, at ng/L levels). Ozone or UV alone has limited effect; the synergistic efficiency is 80–95%.

14.2 Design calculation

Side-stream flow = recirc flow × 5% AOP volume = side-stream flow (m^3/h) × HRT (min) / 60 (m^3) Ozone = side-stream flow × 24 h × 0.5 g/m^3 / 1000 (kg/d)

HRT defaults by AOP type: **integrated** (UV+O_3 single reactor) 2 min; **split** (O_3 contactor + UV reactor, large projects) 8 min. The hydroxyl radical lives <1 ms, so the reaction is photochemically instantaneous; HRT mainly governs contact stability. The AOP unit carries tag **X-901** and the ozone generator **G-901**. AOP effluent must pass through activated carbon or UV to destroy residual ozone before returning to the mainline. Not every RAS needs AOP — only species sensitive to off-flavour.

Chapter 15 Oxygenation and Degassing

15.1 DO saturation (Benson–Krause 1984)

DO_sat = exp(−139.34411 + 1.575701e5/Tk − 6.642308e7/Tk^2 + …)

Temp (°C)	Freshwater DO_sat	Seawater 30‰	Reduction
10	11.29	9.32	−17%
15	10.08	8.39	−17%
20	9.09	7.62	−19%
25	8.26	6.97	−18%
30	7.56	6.41	−18%

15.2 Pure oxygen versus air

Parameter	Pure oxygen (LOX + cone)	Air (Roots blower + diffuser)
DO target	110–165% saturation (by species)	~95% saturation
Suitable density	≥ 40 kg/m^3	< 40 kg/m^3
Absorption efficiency	Cone 80–90%	Fine-bubble SOTE 18%
Operating cost	LOX 0.4 EUR/kg	Electricity only

Target DO saturation by species: salmon 165%, warm-water marine 140%, warm-water freshwater 110–125%, tilapia/shrimp 100–115%.

15.3 Tank DO calculation

Tank outlet DO = DO_sat × satTarget% − deltaDO_peak deltaDO_peak = o2FishPeakHourly / flowM3h × 1000 (mg/L)

The hourly peak is used to assess the DO drawdown, reflecting the post-feeding metabolic spike.

15.4 CO_2 stripper

15.4.1 Design parameters

Parameter	Default	Basis / industry data
Gas:liquid ratio G:L	3	Mid-size projects 2–4, salmon plants 5–7
Air pressure	2.5 kPa (adjustable)	Delta Cooling Towers measured 0.87–1.12 kPa
Blower efficiency	0.60	Roots 0.5–0.7, centrifugal 0.65–0.8
Single-pass removal	65%	G:L = 3, shallow column

15.4.2 Blower power

Airflow (m^3/h) = recirc flow × G:L Blower power (kW) = airflow/3600 × pressure(Pa) / blower efficiency / 1000

Example (salmon grow-out, Q = 10155 m^3/h): 10155 × 3 / 3600 × 2500 / 0.6 / 1000 = 35 kW. The stripper carries tag **C-501**.

15.4.3 Steady-state CO_2 concentration

CO_2_tank (mg/L) = CO_2_Daily (kg/d) × 10^6 / (flow (m^3/h) × 24 × η_co2 × 1000)

The CO_2 limit is species- and salinity-aware (salmonids 15 mg/L; other freshwater 20; seawater 15 — see §3.2 and §4.5). When the stripper is disabled, η_co2 falls back to co2FallbackEff (default 5%).

15.5 Oxygenation-cone topology: mainline vs bypass

International large-scale RAS projects are “more often plumbed in a side-stream (bypass) configuration” (Linde SOLVOX cone, Pentair AES Speece, PR Aqua PPC, Global Seafood Advocate 2019). iRAS models both topologies explicitly.

15.5.1 Physical comparison

Item	Mainline (in-series)	Bypass (high-pressure side-stream)
Mainline path	All recirc water passes the cone	Mainline skips the cone; 5–30% bypass passes a high-pressure cone and returns
Main-pump extra head	+2–15 m (cone loss)	0 (main pump does not pass

Item	Mainline (in-series)	Bypass (high-pressure side-stream)
		the cone)
Cone-rated flow	= full recirc flow Q	= Q × bypassRatio (small flow)
Cone operating pressure	Low 10–21 psi (LHO)	High 1–3 bar (SOLVOX / PPC)
Typical DO out of cone	130–170% saturation	200–400% saturation
Extra equipment	None	Bypass pump P-602 + mixing tee M-601
Suitable scale	Small/medium (Q < 5000 m^3/h)	Large (Q > 5000 m^3/h)

15.5.2 Field definitions

Field	Default	Clamp range
o2ConeTopology	mainline	mainline / bypass
o2ConeHeadLoss_m	3 m	[0, 15]
o2BypassRatio (%)	10	[5, 30]
o2BypassPumpHead_m	20 m	[5, 40]
o2BypassPumpEta (%)	60	[30, 85]

15.5.3 Main-pump head

mainline: H_pump_total = H_main + o2ConeHeadLoss bypass: H_pump_total = H_main (main pump skips the cone; bypass pump pressurises separately)

15.5.4 Bypass pump P-602

Q_bypass = Q × o2BypassRatio P_bypass(kW) = Q_bypass × ρ × g × o2BypassPumpHead_m / (3.6 × 10^6 × o2BypassPumpEta)

P-602 is added to the equipment schedule, CAPEX, OPEX and heat balance, modular with 1+1 redundancy (same convention as the main pump P-101).

Topology selection For salmon/trout projects above 1000 t/yr, the bypass topology is recommended: it shrinks the cone-rated flow ~10×, lowers the main-pump head, and reduces both CAPEX and pumping power. For small/medium projects (Q < 5000 m^3/h) the mainline topology is simpler, because the scale effect of the high-pressure bypass cone only emerges above a threshold project size.

15.6 Air-mode oxygenation: tank aeration blower K-101

In air mode (air aeration) the tank main oxygenation is supplied by a dedicated Roots blower with micro-pore diffusers — essential for warm-water species (tilapia, eel, shrimp), which are the most common large-scale RAS scenario in many markets.

Parameter	Typical value	Notes
Equipment type	Roots blower + micro-pore diffuser	SOTE 18–22% (Wagner 2010)
Airflow basis	o2FishPeakHourly × safety / SOTE	Back-calculated from tank peak O_2 demand
Air pressure	15–25 kPa	Diffuser submergence + pipe resistance
Blower efficiency	60%	Typical Roots blower
Redundancy	1+1 standby	Tank oxygenation failure suffocates fish — dual rotation mandatory
P&ID tag	K-101	Distinct from K-302 (BF blower)

Airflow Q_air (Nm^3/min) = o2FishPeakHourly × safety / (60 × SOTE × 0.28) Power P_K101 (kW) = Q_air/60 × P_kPa × 1000 / 0.60 / 1000

Chapter 16 Heat Load and Temperature Control

16.1 Project-level constant management

All heat-balance parameters (air temperature, source-water temperature, U-values, air-change rate) are project-level constants and live in a top-level global configuration (globalThermalConfig) that applies to all stages. See Chapter 28 and Appendix B.

16.2 Make-up water temperature difference

Make-up volume = V_total × daily exchange rate (m^3/d) Heat load = V × 1000 × 4.18 × ΔT / 3600 (kWh/d)

16.3 Plate heat-exchanger recovery

T_after = T_source + η_HX × (T_target − T_source) η_HX = 75% (iRAS default)

16.4 Heat-pump top-up

Heating and cooling COP are separated: COP_heat default 4.0, COP_cool default 3.0.

16.5 Two-node steady-state solution (overview)

The shop air is heated by the tanks to 5–12°C above ambient, so tank heat loss is smaller than "loss to the outdoor air", and humidity build-up reduces the evaporation rate. iRAS therefore uses a two-node steady-state model solving the water energy balance, the air sensible-heat balance and the air moisture balance simultaneously. The evaporative latent term Q_evap × λ is applied on the **water** node (evaporation removes sensible heat from the water); the air node receives only the moisture increment. The detailed model and solver are in Chapter 29.

16.6 Heat-pump types

Type	COP_heat	COP_cool	Source temp	Use
Air-source	3.5–4.0	2.5–3.0	T_air	Standard, air > −10°C
Water-source	4.5–5.5	3.5–4.5	T_source	Abundant groundwater / river
Ground-source	4.0–5.0	3.5–4.5	T_ground (≈ annual mean)	Has space for buried loops
Open seawater	4.5–5.5	3.5–4.5	T_seawater	Coastal, titanium-plate HX

Engineering notes: in cold climates with warm-water species the temperature lift can be large and heat-pump power may exceed oxygenation power; plate exchangers foul with seawater scale and need acid washing every 3–6 months; insulated piping/tanks cut 20–30% of heat loss; a sealed tank cover (factor none=1.0 / partial=0.4 / full=0.1) can save 20–40% of heating power in severe climates.

Chapter 17 Pumps and Piping Design

17.1 Pump power formula

$$P\ (\mathrm{kW}) = 9.81 \times Q\ (\mathrm{m^3/h}) \times H\ (\mathrm{m})\ /\ (3600 \times \eta)$$

17.2 Pump set

The main recirculation pump usually accounts for 30–40% of total plant electricity and is the single largest consumer; a VFD saves 15–25%. The RDF backwash pump is **not** a separate item (it ships integral to the drum filter). In the bypass oxygenation topology the branch pumps additionally include the cone bypass pump P-602.

Pump	Flow	Default head (m)	Default η	Tag / notes
Main recirculation pump	recirc flow Q	12	68%	P-101, full flow, largest power, VFD
Denitrification branch	Q × denitri ratio	10	60%	P-601
UV branch	Q × UV ratio	10	60%	P-401
AOP branch	Q × AOP ratio	10	60%	P-701

Pump	Flow	Default head (m)	Default η	Tag / notes
Skimmer branch	Q × skimmer ratio	6	60%	P-501, short distance
O_2 cone bypass pump	Q × bypassRatio	20	60%	P-602 (bypass topology only)

17.3 Pipe-diameter selection

Pipe type	Recommended velocity (m/s)	Notes
Suction	0.8–1.2	Too high → cavitation
Discharge	1.5–2.5	Too high → water hammer
Sludge/drain	1.0–2.0	Self-cleaning velocity to prevent settling
Overflow	0.5–1.0	Gravity flow

Pipe diameter $D = \sqrt{(4Q / \pi v)}$ (Q in m^3/s, v in m/s, D in m)

17.4 Head loss

Component	Typical value	Notes
Static head	0.5–2 m	Tank-to-treatment elevation difference
Pipe friction	2–4 m	Set by length and diameter
Equipment resistance	3–5 m	RDF + BF + UV, ~1–2 m each
Safety margin	1–2 m	10–20%
Total	**8–15 m**	**iRAS default 12 m**

Chapter 18 Electrical and Control Systems

18.1 Online water-quality monitoring

Parameter	Sensor	Accuracy	Location
DO	Optical (luminescent)	±0.1 mg/L	Tank + BF outlet
pH	Glass electrode	±0.01	Tank + BF
Temperature	Pt100	±0.1°C	Tank + source water

Parameter	Sensor	Accuracy	Location
TAN	Ion-selective	±0.1 mg/L	Tank (optional)
ORP	Platinum electrode	±5 mV	Skimmer (ozone control)
Level	Ultrasonic	±1 cm	Tank / sump

18.2 PLC control logic

- Main recirculation pump: VFD + PID on level/flow
- RDF backwash: timed or differential-pressure triggered
- Oxygenation: PID on the LOX flow valve
- Alkalinity dosing: PID on the $NaHCO_3$ metering pump
- Methanol dosing: controlled by NO_3
- Temperature: PID heat-pump start/stop
- Make-up: level-triggered

18.3 Alarm system

Alarm	Threshold	Response
Low DO	< 5 (level 1) / < 3 (emergency)	Start standby blower / O_2 cylinder
Low pH	< 6.5	Dose $NaHCO_3$; reduce feeding
High TAN	> 2.0 mg/L	Reduce feeding; check BF
Temperature anomaly	±2°C deviation	Check heat pump
Power outage	Mains interruption	Diesel genset starts in 30 s
Pump fault	Abnormal current	Switch to standby pump

Essentials: a diesel genset rated for 100% plant load; a UPS for PLC and sensors; SMS alarms; and a monthly emergency-switchover drill.

Chapter 19 Biosecurity and Disinfection Strategy

The closed nature of a RAS gives inherently high biosecurity, but once a pathogen enters, the recirculating water spreads it rapidly. “Keeping it out” matters more than “treating it”. New projects should establish a complete biosecurity SOP before stocking.

19.1 Intake disinfection

- Disinfect new source water before it enters the system (UV 60 mJ/cm^2 or ozone 0.5 mg/L residual for 5 min).

- Well water is relatively safe; river/sea water is higher-risk and needs multi-stage treatment.
- Physically separate make-up piping from culture piping to prevent siphoning.

19.2 Personnel and tools

- Change into work clothes and boots, pass through a disinfectant foot-bath (200 ppm sodium hypochlorite).
- Do not share tools between stages; use colour coding.
- Isolate the feed store from the culture hall.

19.3 Quarantine

- Quarantine new fingerlings for 2–4 weeks.
- Give the quarantine tank an independent water-treatment system.
- Isolate sick fish immediately; bury or incinerate mortalities.

19.4 Routine disinfection

Object	Disinfectant	Concentration / method	Frequency
Tools	Iodophor	100 ppm soak 10 min	After each use
Foot-bath	Sodium hypochlorite	200 ppm, change weekly	Daily
Hall floor	Hydrogen peroxide	3% spray	Weekly
Empty tank	Potassium permanganate	50 ppm soak 24 h	Between batches
Intake	UV	60 mJ/cm^2	Continuous

Chapter 20 System Start-up and Commissioning

A new RAS reaches full load over 8–12 weeks. Rushing to full stocking is the most common beginner error — the biofilter capacity is built up gradually, and exceeding it means a TAN spike and mass mortality.

20.1 Start-up sequence

Phase	Time	Action	Key indicator
Equipment commissioning	Week 0–1	No-load run; pressure test	48 h fault-free
BF seeding	Week 1–4	Inoculate; NH_4Cl to TAN 2–4	TAN falls, NO_2 appears

Phase	Time	Action	Key indicator
Nitrification maturity	Week 4–6	Continue seeding	TAN<0.5 & NO_2<0.3 for 3 d
Low-load stocking	Week 6–8	Stock to 20–30% capacity	TAN<1.0 stable
Ramp load	Week 8–12	+15–20% per week	Indicators safe
Full load	Week 12+	Normal production	TAN<1, NO_2<0.3, DO>6

20.2 Common start-up problems

Problem	Cause	Action
TAN not falling	Temperature / pH too low	Raise to 25–30°C; dose alkalinity
NO_2 persistently high	Nitrobacter not established	Normal — wait patiently
TAN spike after stocking	Too many fish / too much feed	Cut feeding 50%; partial exchange
Rapid pH drop	Alkalinity exhausted	Dose $NaHCO_3$
Fish not feeding	Stress	Check water quality; reduce feeding

PART IV

Economic Evaluation

Chapter 21 Operating Cost

21.1 Cost structure

Cost item	Basis	Typical share
Feed	daily feed × feed price	50–65%
Electricity (pumps)	main + branch pumps × 24 h	8–15%
Electricity (BF blower)	BF coarse aeration × 24 h	3–5%
Electricity (UV)	UV power × 24 h	1–3%
Electricity (heat pump)	heat-pump energy × power price	0–15%
Liquid oxygen	o2Fish / absorption × unit price	5–12%
Ozone power	ozone generator × 24 h	1–3%
CO_2 blower power	stripper blower × 24 h	2–5%
Methanol	denitrification × unit	1–3%

Cost item	Basis	Typical share
	price	
$NaHCO_3$	alkalinity dose / 0.595 × unit price	1–3%
Misc. electricity	lighting / control / standby	1–2%
CAPEX depreciation	equipment/8 yr + civil/20 yr / 365	15–30%

Consumable unit prices (iRAS v1.8 defaults): electricity 0.10 EUR/kWh, liquid oxygen 0.4 EUR/kg, methanol 0.6 EUR/kg, $NaHCO_3$ 0.4 EUR/kg.

21.2 Peak versus daily-average

Peak case is based on feedMax (= biomass × feedRate); the daily-average case on feedAvg (= annual growth × FCR / 365). Peak cost is typically 1.5–2.5× the daily average. Equipment must withstand the peak, but day-to-day operation runs near the average. Chemical/ozone consumption in the annualised case is scaled by feedRatio (= feedAvg/feedMax).

21.3 CAPEX depreciation

iRAS uses dual depreciation lives:

- **Equipment** (default 8 yr, adjustable 5–20): drum filters, pumps, blowers, UV, ozone, etc.
- **Civil works** (default 20 yr, adjustable 10–30): tank shells, piping, installation, commissioning, design.

Annual depreciation = equipment / equip life + civil / civil life (EUR/yr) Daily depreciation = annual depreciation / 365

Chapter 22 Investment Estimate

22.1 Direct equipment cost (v1.8 unit prices, EUR)

Equipment	Unit	Default price	Recommended range
Drum filter (RDF)	EUR/(m^3/h)	130	78–208
MBBR system (media + aeration)	EUR/m^3 media	1,169	779–1,558
UV disinfection	EUR/kW	2,597	2,078–3,896
Protein skimmer	EUR/m^3	6,494	3,896–10,390
Ozone generator	EUR/(kg/d)	25,974	20,779–38,961
AOP reactor	EUR/m^3	9,091	5,195–12,987

Equipment	Unit	Default price	Recommended range
Denitrification reactor	EUR/m^3	1,429	1,039–2,078
CO_2 stripper	EUR/m^2	5,195	3,896–7,792
Main pump + VFD	EUR/kW	909	649–1,299
Branch pump (each)	EUR/kW	779	519–1,039
Roots blower	EUR/kW	1,039	779–1,558
LOX oxygenation tower	EUR/(m^3/h)	91	52–130
Heat pump (air-source)	EUR/kW	519	390–779
Plate heat exchanger	EUR/m^2	312	208–519
LOX tank (site equipment)	EUR/tank	100,000	site-configured × count

The LOX storage tank is configured as **site-level equipment** (siteEquipConfig.oxygenTank: size, count, unit price), not split per stage.

22.2 Total project investment

Total project investment = direct equipment × project multiplier (Lang factor) project multiplier = 2.2 (default, range 1.8–2.5)

The multiplier covers piping & valves (15–20%), electrical & instrumentation (10–15%), control (5–10%), installation & commissioning (10–15%), civil works & buildings (30–40%) and design & supervision (5–8%). Estimate accuracy is about ±40%; actual cost depends on vendor quotations.

22.3 Scale and regional factors

Two further multipliers are applied to the unit prices, so the same design scales realistically across project sizes and regions:

- **Scale-effect (Lang) exponent** (default **0.10**, baseline 1000 t/yr): larger projects amortise automation, design and management over more output, lowering unit CAPEX. The factor equals 1.0 at the baseline scale, falls below 1.0 above it, and rises above 1.0 below it. An exponent of 0 disables the effect (linear CAPEX).
- **Regional factor** (default **1.0**, range 0.3–1.5): adjusts unit prices for the local cost level (labour, logistics, import duties).

22.4 Effect of cone topology on CAPEX

The oxygenation-cone topology has a marked effect on CAPEX. Relative to mainline, the bypass topology shrinks the LOX-tower rated flow ~10× (priced on the small bypass flow) and lowers the main-pump head, while adding a small bypass pump P-602. The net effect for large salmon projects is a lower total project investment, even after the added bypass pump — the saving on the oxygenation tower dominates.

PART V

Design Cases

Chapter 23 Case: Atlantic Salmon, 1000 t/yr

This case uses the iRAS default Atlantic-salmon parameters and a Norwegian Bergen winter climate (T_air = 0°C, T_source = 4°C, T_target = 15°C), the most representative climate for real salmon projects.

23.1 Stage parameters

Parameter	Fry	Juvenile	Grow-out (mid)	Market size
Weight (g)	0.1→50	50→500	500→2000	2000→5000
Months	6	8	6	4
FCR	1.0	1.1	1.2	1.20
Protein (%)	50	45	42	40
feedRate (%)	3.0	1.5	1.0	0.95
Temp (°C)	12	14	15	15
Density (kg/m^3)	30	50	60	70
Salinity (‰)	0	0	12	30
Loss (%)	15	8	4	3
recDepth (m)	1.5	3.0	5.0	7.0
Modules	1	2	3	4

23.2 Key results

Item	Fry	Juvenile	Grow-out	Market	Whole farm
Standing stock (t)	3	41	132	237	413
V_tank (m^3)	111	836	2,193	3,385	6,525
V_total (m^3)	~200	~1,305	~3,208	~5,183	~9,896
Recirc flow Q (m^3/h)	424	2,465	5,481	10,155	—
feedMax (kg/d)	~96	~616	~1,316	~2,251	~4,279
Total O_2 demand (kg/d)	49	308	662	1,113	~2,132

Item	Fry	Juvenile	Grow-out	Market	Whole farm
Reverse pool dia. D (m)	10	9.4	10.5	8.8	—

Design-basis stage The whole-farm column sums the four stages, but the market-size stage (the highest flow and oxygen demand) is normally the design basis. For this case the market-size totals — o2DailyTotal ~1113 kg/d, peak cone supply ~41.2 kg/h, hourly peak O_2 ~69.6 kg/h — are the key sizing figures.

23.3 Topology choice

For this 1000 t salmon case the bypass cone topology is the recommended default: it lowers the total project investment (mainly through the smaller oxygenation tower and reduced main-pump head) and the annual pumping power, which raises the project IRR and shortens the payback relative to the mainline topology. This matches the international engineering practice of plumbing large salmon RAS oxygenation cones in side-stream configuration.

Chapter 24 Case: Turbot, 500 t/yr

24.1 Stage parameters

Parameter	Fry	Grow-out (mid)	Market size
Weight (g)	3→50	50→250	250→600
Months	4	8	6
FCR	0.9	1.1	1.3
Protein (%)	55	50	48
feedRate (%)	3.0	1.5	0.9
Temp (°C)	16	16	16
Density (kg/m^3)	25	50	60
Salinity (‰)	28	28	28
recDepth (m)	0.8	1.0	1.0

Turbot is a benthic flatfish; water depth is kept within ~1 m and biomass is held mainly by floor area. The whole-farm standing stock is ~809 t and V_tank ~5,121 m^3. iRAS turbot modelling agrees with industry survey data to within a few percent, reaching the accuracy needed for engineering estimates.

Chapter 25 Case: Whiteleg Shrimp, 100 t/yr

25.1 Clear-water RAS versus biofloc

Dimension	Clear-water RAS	Biofloc (BFT)
Treatment core	Drum filter + biofilter	In-situ heterotroph flocs treat TAN
Biofilter	Yes (MBBR)	No (the key difference)
Water colour	Clear	Brown (flocs 300–800 mg/L)
C:N ratio	No requirement	C:N ≥ 15:1 critical
Density	2–15 kg/m^3	3–8 kg/m^3
iRAS support	Yes	No

25.2 Stage parameters

Parameter	Fry	Grow-out early	Grow-out late
Weight (g)	0.01→1	1→10	10→30
Months	1	1.5	1.5
FCR	1.0	1.3	1.6
Protein (%)	45	42	38
feedRate (%)	10	5	3
Temp (°C)	30	30	30
Density (kg/m^3)	2	8	15
Salinity (‰)	15	15	15
o2FishFactor (30°C)	0.40	0.40	0.40

Shrimp Q_{10} is set to 1.3 (versus 2.0 for fish), reflecting metabolic saturation at high temperature (30°C). The whole-farm steady-state standing stock is ~6.25 t and V_tank ~558 m^3. Small shrimp projects show a higher unit cost because iRAS does not by default apply a strong CAPEX scale exponent; raising the scale (or the scale exponent) brings the unit cost down.

PART VI

Platform and Engineering Features

Chapter 26 Multi-Module Parallelism and N+1 Redundancy

26.1 Why large projects are modular

For projects above 1000 t/yr, engineers do not build one huge loop but split it into several parallel modules: fault isolation (one module can be serviced while others run), sensible equipment sizes (a single main pump stays below 75–90 kW and flow below ~600 m^3/h), phased construction, standardised maintenance and dispersed biosecurity risk.

Scale	Typical modules	Per-module capacity
< 500 t/yr	1	Whole farm in 1 module
500–1500 t/yr	2–3	200–500 t/yr per module
1500–5000 t/yr	3–6	500–1000 t/yr per module
> 5000 t/yr	5+	1000+ t/yr per module

26.2 iRAS modular logic

Each stage's module count is set by the user (subSystemCount, default 1, range 1–10). Different stages may use different module counts — e.g. fry 1, juvenile 2, grow-out 3, market 4 — reflecting the increasing need to disperse risk as production grows:

numModules = stage.subSystemCount per-module flow = stage total flow / numModules per-module capacity = stage total equipment capacity / numModules

26.3 In-module N+1 redundancy

Equipment	Single-unit cap	Sizing rule	Standby
Main pump P-101	75 kW	ceil(module flow power / 75)	+1
BF blower K-302	30 kW	ceil(BF blower power / 30)	+1
RDF F-201	600 m^3/h	ceil(module flow / 600)	+1
UV D-401	8 kW	2 in parallel, 75% margin each	1 fails → other ramps up
Heat pump H-801	200 kW	ceil(heat-pump kW / 200)	+1
Branch pumps	30 kW	ceil(branch power / 30)	+1
Ozone generator G-901	50 kW	ceil(O_3 power / 50)	+1

26.4 Shared site-level equipment

Equipment	Configuration	Notes
LOX tank V-901	2 tanks (duty + standby)	Dual-tank, switch without stopping
Emergency genset	1 × 100% plant load	30 s start + auto-transfer
Transformer	2 sets (HV/LV dual)	Fault transfer
UPS	1 set (PLC + sensors + key valves)	10–30 min ride-through

Chapter 27 Tank Configuration and Unified Depth

27.1 Shape and size logic

Circular tanks (Cornell dual-drain) are the most common because the tangential inflow creates a self-circulating vortex that exercises the fish, solids settle toward the centre for efficient bottom-drain collection, and there are no dead corners. Key dimensions: diameter D = 4–30 m, depth H = 1–8 m (by species), height-to-diameter ratio H/D = 0.15–0.5, single-tank volume 50–3000 m^3.

27.2 Back-calculation

total tanks = poolsPerSet × numModules single-tank V = V_tank / total tanks reverse diameter D = √(4V / πH) H/D ratio = H / D

27.3 Automatic engineering warnings

Condition	Warning
D < 4 m	Tank diameter too small — reduce pools per set or parallel sets
D > 30 m	Tank diameter too large — increase pools per set or parallel sets
D > 20 m and H < 3 m	Large shallow tank (H/D < 0.15), prone to dead zones — deepen
poolsPerSet = 1	Only 1 tank per set — no duty/standby rotation; convention is ≥ 2

27.4 Recommended depth by species (recDepth_m)

Species	Fry	Juvenile	Grow-out	Market	Rationale
Salmon	1.5	3.0	5.0	7.0	Pelagic swimmer, deep tanks (commercial 7–12 m)
Turbot	0.8	—	1.0	1.0	Benthic, shallow single-layer floor area
Grouper	1.2	—	2.5	3.0	Semi-benthic, medium depth
Largemouth bass / mandarin	1.5	2.5	3.0	—	Freshwater carnivore, water column

Species	Fry	Juvenile	Grow-out	Market	Rationale
Tilapia	1.0	2.0	3.0	—	Tolerates high density
Eel	0.8	1.5	2.0	—	Benthic, medium-shallow
Shrimp	1.5	2.0	2.5	—	Benthic + water column, medium

27.5 Unified depth data source

Tank depth uses a single value with a clear priority, shared by the pool-diameter back-calculation and the heat-balance solver: (1) user-entered stage depth, (2) species-recommended recDepth_m from the species database, (3) proc.poolDepth when present, (4) 1.5 m default. Using one depth source ensures the tank geometry and the heat model are always consistent.

Chapter 28 Global Climate and Envelope Modelling

28.1 Project-level constant management

Heat-balance parameters are project-level constants (the same building serves every stage), so they live in a top-level global configuration set once and used by all stages: air temperature (winter/summer/annual), source-water temperature, relative humidity, U-values (wall, roof, floor, window), window-to-wall ratio, air-change rate and heat-recovery efficiency.

28.2 Climate presets

Region	Winter air	Winter source	Summer air	Summer source	Annual
Qingdao (coastal)	−7°C	4°C	32°C	24°C	13°C
Hainan	17°C	22°C	33°C	30°C	25°C
Norway (Bergen)	0°C	4°C	17°C	13°C	8°C
Faroe Islands	2°C	6°C	13°C	11°C	7°C
Northern inland	−15°C	2°C	30°C	20°C	8°C
South China	10°C	15°C	33°C	26°C	22°C

28.3 Envelope presets

Preset	U_wall	U_roof	U_floor	U_window	Notes
Industrial	0.8	0.6	0.8	2.5	50–80 mm

Preset	U_wall	U_roof	U_floor	U_window	Notes
standard					PU insulation, double glazing
Nordic premium	0.25	0.20	0.30	1.2	200 mm mineral wool, triple Low-E
Basic shed	2.0	1.5	1.5	5.0	Single-skin steel, single glazing

28.4 Three-condition comparison

Every calculation runs three conditions simultaneously: **design winter** (coldest-month electricity peak, sizes the heat pump), **design summer** (hottest-month peak, sizes the chiller) and **annual mean** (annualised electricity for costing). The installed heat-pump capacity takes the larger of the winter/summer extremes.

28.5 Tank-cover type (evaporation factor)

Cover type	Factor	Typical evaporation	Use
None	1.0	Full open-surface	Outdoor or basic shed
Partial	0.40	~40%	Standard industrial RAS
Full (sealed)	0.10	~10%	Nordic / severe climate, energy-saving

Evaporation is latent heat (2257 kJ/kg) and is one of the largest winter heat-loss terms; a sealed cover can save 20–40% of heating power.

28.6 Ventilation and heat recovery

$$\text{Ventilation sensible loss} = 0.34 \times V\_room \times ACH \times (1 - ventHeatRecovery) \times (T\_room - T\_air)$$
$$0.34 = \rho\_air \times c\_p,air / 3600 = 1.2 \times 1005 / 3600$$

ACH = 0.3–0.5 for Nordic high-insulation halls; 1.0–2.0 for standard halls; 3.0+ for basic or high-humidity halls. Heat recovery (plate or rotary) recovers 70–85% of the sensible heat; Nordic projects typically use 0.7–0.85, others 0.

Chapter 29 Two-Node Steady-State Heat Balance

29.1 Why two nodes

A single-point balance assumes the tank loses heat to an infinite outdoor environment. In reality the shop air is heated by the tanks to 5–12°C above ambient (so tank loss is smaller than loss to outdoor air), humidity build-up lowers the evaporation rate, winter condensation on cold surfaces recovers water, and in summer cooling the air can warm the tanks (reverse flow). The two-node model captures these effects.

29.2 Two-node steady-state equations

```
Node 1 — water energy balance (kWh/d): Q_metabolism + Q_equipment + Q_makeup − Q_pool_loss − Q_evap × λ = Q_HP Q_pool_loss = UA_pool × (T_w − T_room) × 24/1000 Q_evap × λ = evaporative latent loss (on the WATER node) Node 2 — air sensible-heat balance (W): UA_pool × (T_w − T_room) + Q_air_gain = UA_other × (T_room − T_air) + UA_floor × (T_room − T_ground) + UA_vent × (T_room − T_air) Node 3 — air moisture balance (kg/s): m_evap × cover × A_water = m_air × ACH × (W_room − W_outdoor)
```

The evaporative latent term Q_evap × λ sits on the water node (evaporation removes sensible heat from the water as molecules leave); the air node receives only the moisture increment, not the λ term directly. The nodes are coupled (T_room sets W_sat; W_room sets m_evap) and solved by Picard iteration (the implementation uses a closed-form solution that typically converges in one pass).

29.3 Evaporation/condensation model

```
m_evap [kg/(m²·s)] = β × (W_sat(T_w) − W_room) β = h_c × Lewis_factor / c_p_air ≈ 2–5 × 10⁻⁴ kg/(m²·s)
```

When W_sat(T_w) > W_room evaporation occurs (m_evap > 0, removing heat from the water); when W_sat(T_w) < W_room condensation occurs (m_evap < 0, releasing heat to the water). Condensation is typical in a hot-humid climate culturing a cold-water species; the condensate is credited against the make-up volume, reflecting the water-saving benefit.

29.4 Solver outputs

Output	Meaning	Typical range
T_room	Shop air temperature	T_air + 5 to +12°C
RH_room	Shop relative humidity	55–80%
m_evap	Evaporation rate (negative = condensation)	±5 to ±50 kg/h
Q_envelope	Total envelope heat loss	5–50 kW
Q_pool	Tank heat loss	5–30 kW
heatPumpKWhPerDay	Heat-pump daily energy	100–3000 kWh/d
cop	COP for the current	2.5–5.5

Output	Meaning	Typical range
	condition	

29.5 Humidity warning

When shop RH exceeds 85%, iRAS flags it and suggests: raise ACH (at the cost of more ventilation heating), add a sealed tank cover (reduce the evaporation source), install a dedicated dehumidifier, or add exhaust heat recovery (recover waste heat while dehumidifying).

Chapter 30 Automatic P&ID Generation

30.1 What a P&ID is

A Piping & Instrumentation Diagram is the core design document describing process units, pipe connections, flow direction, key instruments and control logic. A complete P&ID includes all major equipment (with tags), pipes (line weight for main/branch, colour for medium), valves and fittings, instrument bubbles, control interlock lines and a title block.

30.2 iRAS P&ID generation

iRAS provides a standalone P&ID viewer (pid.html), opened from the main page. Data flow: (1) after a calculation the main page serialises the current design to JSON in localStorage; (2) the new window opens pid.html and reads the data on load; (3) a repeat click notifies the open window to refresh via postMessage.

30.3 Layout

The P&ID uses a two-layer layout. Mainline (upper): T-101 fish tank → F-201 drum filter → B-301 MBBR → C-501 CO_2 stripper → S-101 sump → P-101 main pump → A-601 oxygenation cone → back to T-101. Side-streams branch from mainline tees (S-701 skimmer, R-801 denitrification, D-401 UV). Utility layer (lower): V-901 LOX tank, G-901 ozone generator, X-901 AOP, K-302 BF blower, H-801 heat pump.

30.4 Auto-generated equipment table

Tag	Name	Example spec (market stage)
T-101	Fish tank	8 tanks × Φ8.8 × 7 m, 423 m^3/tank
F-201	Drum filter	Q 11170 m^3/h, η_TSS 75%
B-301	MBBR biofilter	V 438 m^3, η_TAN 70%, fill 50%
C-501	CO_2 stripper	G:L 3, η 65%
P-101	Main pump	488 kW @ 12 m, 68% η,

Tag	Name	Example spec (market stage)
		VFD
A-601	Oxygenation cone	41.2 kg O_2/h, η 85%
S-701	Protein skimmer (side)	α 20%, V 74 m^3
D-401	UV (side)	α 10%, 38 kW @ 40 mJ/cm^2
R-801	Denitrification (side)	α 10%, V 129 m^3
V-901	LOX tank	1169 kg/d (market), 3 t tank
X-901	AOP (utility)	α 5%, V 17 m^3

30.5 Interlock examples

Main pump P-101 is controlled by tank DO-101 through a PID loop on the VFD frequency, holding tank DO ≥ the species minimum saturation. Other interlocks: pH-102 → $NaHCO_3$ dosing pump; TT-103 → H-801 heat pump; LT-104 → make-up valve; AT-302 (BF outlet TAN) → feeding-reduction alarm; DO-601 (cone outlet DO) → LOX flow valve.

30.6 Export

pid.html supports direct SVG printing (browser Ctrl+P → save as PDF) or SVG file download. The SVG vector format scales losslessly and can be imported into AutoCAD/Visio as an engineering base drawing.

Chapter 31 Equipment Schedule Export

31.1 Engineering uses

The equipment schedule is a core deliverable, used for vendor quotation, procurement contracts, acceptance, and maintenance/spares planning.

31.2 Structure

iRAS provides a standalone equipment-list page (equipment.html), opened from the main page. It contains: project metadata (species / annual yield / save time); a separate list per stage; under each stage the modular note (N modules), tank configuration, mainline equipment, side-stream equipment and utilities, plus a stage subtotal; and a whole-farm total.

31.3 N+1 expansion

Each item is expanded into duty + standby. For example, a market stage with 4 modules where each module needs 2 main pumps plus 1 standby yields 12 main pumps in total (8 duty + 4 standby).

31.4 Unit price and amount

item subtotal = quantity × unit capacity × unit price (EUR/capacity) direct equipment = sum of item subtotals total project cost = direct equipment × project multiplier (default 2.2)

31.5 Single-listed static equipment

Some high-value items need no N+1 standby but a single unit with a bypass valve: the MBBR biofilter (media buffers for hours), the denitrification reactor (NO_3 accumulates slowly), the protein skimmer (slow DOM accumulation), the CO_2 stripper column (its blowers have N+1, the column itself has no moving parts) and the LOX tank (duty + standby dual-tank). These are marked "1 set (single + bypass valve)" and are not expanded into multiple units.

31.6 Data transfer

The main page and equipment.html share data via localStorage (key iRAS_equipmentExport). The main page writes the full export object on each click; the new window reads it on load; a repeat click notifies the open window to refresh. The P&ID and equipment-list keys are kept separate (iRAS_pidExport and iRAS_equipmentExport) so the two windows can stay open simultaneously without interfering.

Chapter 32 Multi-Condition Cost Baseline

32.1 Why compare conditions

Siting determines climate and water temperature, and regional cost differences are large. A standardised multi-condition comparison supports pre-feasibility (comparing regions), seasonal analysis (winter/summer operating cost), design selection (sizing the heat pump for winter or summer) and financing (a cost range).

32.2 Key observations

- **Climate dominates the heat-pump cost:** across conditions, the operating-cost spread is mostly driven by heat-pump energy.
- **Cold-water species in hot climates is expensive:** culturing a 12–15°C species where summer air is ~32°C means a 17–20°C cooling lift and a steep cooling bill — generally not recommended.
- **Mild maritime climates are favourable:** a cool summer avoids heavy cooling, and a cold but stable source water keeps the heating lift modest.

32.3 Calibration against literature

iRAS engineering cost estimates fall within roughly ±10–20% of published industrial figures, which is sufficient for the ±40% accuracy of a feasibility-stage estimate. iRAS excludes "soft costs" (labour, fingerlings, management, insurance), so its all-in cost is typically 20–30% below published whole-cost figures; these should be added separately in practice.

Chapter 33 Investment and Financial Evaluation

33.1 Why financial evaluation

Engineering estimates (CAPEX, annual OPEX, all-in cost per kg) are only the starting point. The real questions are whether the project earns money (IRR), how fast it pays back (static/dynamic payback), whether NPV is positive, what the break-even output is, whether debt leverage amplifies equity returns, and which variable the project is most sensitive to. iRAS provides a standalone evaluation page (finance.html) that automates this.

33.2 Data flow

main page calculateAll → lastResults.summary (financial summary) → buildPIDExport → JSON (summary.finance + perStage[i].cost/capex) → localStorage['iRAS_financeExport'] → finance.html reads → user sets parameters → 8 tables + KPIs + charts → export Excel (.xlsx) or print PDF

33.3 User input parameters

Group	Parameter	Default	Notes
Sales	Sale price (EUR/kg)	by species	ex-works (salmon 8, turbot 11, grouper 18, bass 7, ...)
Sales	Project life (yr)	15	5–30
Sales	Year-1 ramp (%)	30	BF seeding + fry rearing
Sales	Year-2 ramp (%)	70	gradual ramp
Sales	Year-3+ ramp (%)	100	full capacity
Investment	Working capital (% × OPEX)	15	recovered in the final year
Investment	Equity share (%)	30	remaining 70% is bank debt
Investment	Loan rate (%/yr)	5	typical
Investment	Repayment method	equal payment	or equal principal
Investment	Loan term (yr)	8	usually shorter than project life
Evaluation	Discount rate (%/yr)	8	for NPV
Evaluation	Income tax (%)	25	corporate income tax
Evaluation	Sales tax surcharge (%)	0	depends on jurisdiction
Evaluation	Repair rate (% × equip)	2	engineering convention
Evaluation	Admin rate (% × sales)	3	typical

Group	Parameter	Default	Notes
Evaluation	Labour cost (EUR/yr)	by scale	~50,000 EUR/person

33.4 Cash-flow model

Operating CF = sales revenue (revenue × ramp[t]) − sales tax − operating cost (variable × ramp + fixed) − repair (capex_eq × repairRate) − admin (revenue × adminRate) − labour − income tax (max(0, profit) × taxRate) Investing CF: t=0 −fixedAsset; t=1 −workingCapital; t=N +workingCapital + salvage salvage = equipment × 5% + civil × 20% Financing CF (equity view only): t=0 +loan; t=1..loanYears −interest −principal Total-investment CF = Operating + Investing Equity CF = Total-investment CF + Financing

33.5 IRR solver

IRR is the discount rate that makes NPV = 0. iRAS solves it by Newton–Raphson (start $r = 0.10$, tolerance $|\Delta r| < 1\text{e-}7$, max 100 iterations); when Newton–Raphson diverges (e.g. multiple roots or an unusual sign change) it falls back to bisection on $[-0.5, 5.0]$. The two methods together handle the common RAS cash-flow shapes robustly, including a zero-interest loan.

33.6 Eight standard financial tables

#	Table	Main content
1	Investment estimate	Equipment + civil + working capital = total investment
2	Funding & repayment	Equity/debt split + yearly interest/principal
3	Total cost (full-capacity year)	Operating + repair + admin + labour + depreciation + interest
4	Sales revenue & tax	Yearly: ramp → output × price → revenue / tax
5	Profit & loss	Yearly: revenue − costs − tax = net profit
6	Cash flow (total investment)	Net CF + cumulative + discounted → IRR
7	Cash flow (equity)	Includes loan inflow + repayment → equity IRR
8	Sensitivity	5 factors ± effect on IRR, ranked by swing

33.7 Six KPIs

Indicator	Definition	Criterion
Total-investment IRR (after-tax)	IRR of all project cash flows (no debt)	> discount rate (8%) → viable

Indicator	Definition	Criterion
Equity IRR	Shareholder view (project CF + loan − repayment)	leverage effect
NPV	Cumulative CF discounted to t=0	> 0 → viable
Static payback	First year cumulative net CF ≥ 0 (interpolated)	typically 5–8 yr
Dynamic payback	First year cumulative discounted CF ≥ 0	static + discount effect
Break-even point (BEP)	Output ratio where full-year net profit = 0	< 75% → large safety margin

33.8 Sensitivity (5 factors)

Factor	Variation	Typical IRR swing
Sale price	±20%	largest (±5–7%)
CAPEX	±20%	±3–4%
Output	±10%	±1–2%
Feed cost	±20% × feed share	±1–1.5%
Electricity	±20% × elec share	±0.4–0.6%

The feed and electricity shares are read dynamically from the actual OPEX composition (feed share = feed cost / annual OPEX), so the sensitivity table adjusts automatically by species. The discount rate does not affect IRR (IRR is the rate that makes NPV = 0) and is excluded from the table; results are ranked by IRR swing and drawn as a tornado chart. For a RAS the top risk factor is almost always the sale price.

33.9 Species comparison

Using the same scale at different species and prices, the evaluation shows that RAS suits high-value species (> ~8 EUR/kg) with intensive requirements (marine, cold-climate or specialty). Low-value species where pond culture is already cheap rarely justify a RAS — an economic, not a technical, conclusion.

33.10 Excel export

finance.html exports an Excel workbook (SheetJS) with seven sheets: investment estimate, funding + repayment, profit & loss, total-investment cash flow, equity cash flow, sensitivity, and a key-indicators summary. If the CDN library cannot load offline, the page prompts the user to print to PDF instead.

Chapter 34 Feasibility Report Generator

34.1 Purpose

Engineering data (P&ID, equipment schedule, financial Excel) still has to be assembled into a Word feasibility report — the largest piece of manual work between concept and project approval. iRAS provides a report generator (report.html) that maps the engineering and financial data into a structured report template and outputs a .docx document. The user then adds the content that must be filled in manually (project identity, site conditions, detailed civil design), compressing report writing from weeks to days.

34.2 Key design decisions

- **Structured content, not finished layout:** every design firm has its own letterhead/template, so iRAS outputs structured content for the user to paste into their own template rather than imposing a layout.
- **Template + conditional branches, not an LLM:** a feasibility report is a quasi-legal document; determinism and traceability outweigh fluent but unverifiable generated text.
- **nunjucks template engine** with templates embedded in the page, preserving single-file deployment.
- **docx.js** for browser-side .docx generation (lazy-loaded).

34.3 Data flow

main page calculateAll → lastResults.summary → buildPIDExport → JSON (farm + workflow + capex + opex + finance) → localStorage['iRAS_reportExport'] → report.html reads → assembles a multi-namespace context → metadata form (user) → context.project → nunjucks render per chapter → parse to docx object tree → SVG charts (process flow / tornado / cash flow) → PNG → ImageRun → docx Packer → user downloads .docx

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34.4 Data namespaces

Namespace	Source	Main fields
project	user form in report.html	project name / owner / address / design firm / qualifications
farm	iRAS main-page design	species / scale / per-stage configuration / tank back-calc / standing stock / modelled water quality (pH, CO_2, NH_3-N, Ω, β, alkalinity)
workflow	iRAS perStage process data	RDF / BF / UV / CO_2 / cone / skimmer / denitrification / AOP / sump
capex	iRAS investment estimate	direct equipment / civil /

Namespace	Source	Main fields
		working capital / total investment
opex	iRAS operating cost	annual OPEX / all-in cost / depreciation / power / LOX
finance	finance.html logic	params (user inputs) + indicators (KPIs) + tables (8 tables)
meta	report metadata	generation time / iRAS version / template version

34.5 Embedded charts

The report embeds three charts generated browser-side as SVG → PNG: a process-flow schematic (Chapter on technical concept), a cumulative cash-flow chart and a sensitivity tornado chart (financial chapter). For the English report, charts use Latin sans-serif fonts and high-DPR rendering.

PART —

Appendices

Appendix A Symbol Table

Mass balance and concentration

Symbol	Meaning	Unit	Default / range
Δ	Per-cycle pollutant increment	mg/L	—
R	Total pass-through fraction	—	—
C_out	Tank outlet concentration	mg/L	—
C_in	Return-water concentration	mg/L	—
η_rdf	RDF TSS removal	%	75
η_bio	BF TAN nitrification	%	70
VTR_{20}	BF volumetric removal @20°C	$kg/(m^3·d)$	0.5
VDR_{20}	Denitrification @20°C	$kg/(m^3·d)$	0.8
θ_nit / θ_den	Temperature coefficient	—	1.10 / 1.08

Symbol	Meaning	Unit	Default / range
f_sal	Salinity correction	—	max(1−0.01S, 0.3)
SOTE	Oxygen transfer efficiency	%	coarse 8% / fine 18%
tanCoef	TAN coefficient	—	0.092 (fw) / 0.110 (marine carnivore)

Physical framework

Symbol	Meaning	Default
o2FishFactor	Fish respiration factor	base × Q_{10}^((T−20)/10)
o2Q10	Temperature sensitivity	2.0 (fish) / 1.3 (shrimp)
o2DOMfactor	Heterotroph O_2	0.10 (range 0.05–0.20)
peakFactor	Intra-day peak ratio	1.5
safety*	Engineering safety factor	1.10–1.15
COP_heat / COP_cool	Heat pump	4.0 / 3.0
η_HX	Plate HX recovery	75%
depYearsEquip	Equipment depreciation life	8 yr
depYearsCivil	Civil depreciation life	20 yr
V_tank / V_process / V_total	Volumes	biomass/density; sum of units; sum

Investment and finance

Symbol	Meaning	Default
capexFactor	Project multiplier (Lang)	2.2 (1.8–2.5)
scaleExponent	CAPEX scale exponent	0.10 (baseline 1000 t)
regionFactor	Regional cost factor	1.0 (0.3–1.5)
price	Sale price (EUR/kg)	by species
years	Project life	15 (5–30)
ramp1/2/3	Yearly ramp	30 / 70 / 100%
workingCap	Working capital (% × OPEX)	15
ownPct	Equity share	30%
loanRate / loanYears	Loan rate / term	5% / 8 yr
discount	Discount rate (NPV)	8%
taxRate	Income tax	25%
repairRate / adminRate	Repair / admin	2% / 3%
laborCost	Labour	~50,000 EUR/person

Cone topology and discharge

Symbol	Meaning	Default / range
o2ConeTopology	Cone topology	mainline / bypass
o2ConeHeadLoss_m	Cone head loss (mainline)	3 m [0, 15]
o2BypassRatio	Bypass flow ratio	10% [5, 30]
o2BypassPumpHead_m	Bypass pump head	20 m [5, 40]
o2BypassPumpEta	Bypass pump efficiency	60% [30, 85]
sim.discharge.V_m3_d	System discharge flow	V_total × exchangeDaily
sim.discharge.TSS_kg_d	Discharge TSS	load.tssDaily (mass conservation)

Appendix B Global Thermal Configuration

Field	Type	Default	Notes
climateRegion	select	—	Auto-fills T_air / T_source / RH
thermalScenario	select	design_winter	3-condition switch
envelopePreset	select	industrial_standard	Auto-fills U-values
hpType	select	air	4 types
T_air_design_winter/summer/annual	number	—	°C
T_source_design_winter/summer/annual	number	—	°C
T_ground	number	≈ annual mean air	°C
RH_winter/summer/annual	number	—	%
U_wall / U_roof / U_floor / U_window	number	0.8 / 0.6 / 0.8 / 2.5	$W/(m^2 \cdot K)$
windowRatio	number	0.05	Window-to-wall ratio
airChangeRate (ACH)	number	1.0	1/h
ventHeatRecovery	number	0	0–0.85
poolCover	select	partial	none / partial / full
buildingArea_user	number	0 (auto)	0 → auto = 1.6 × pool area
buildingHeight	number	6	m
thermalSharingFactor	number	0.75	Shared-hall heat-loss factor 0.5–1.0

Appendix C Equipment Tag Naming (ISA-5.1)

iRAS follows the ISA-5.1 convention (letter + number). Leading-letter meanings:

Letter	Meaning	Example
T	Tank / pool	T-101 fish tank
F	Filter	F-201 drum filter
B	Biofilter / bioreactor	B-301 MBBR
K	Kompressor / blower	K-101 tank aeration blower, K-302 BF blower
D	Disinfection	D-401 UV
C	Column	C-501 CO_2 stripper
S	Separator / sump	S-101 sump, S-701 skimmer
A	Aerator / absorber	A-601 oxygenation cone
R	Reactor	R-801 denitrification
X	eXtra / advanced	X-901 AOP
G	Generator	G-901 ozone
V	Vessel / storage	V-901 LOX tank
H	Heat	H-801 heat pump
P	Pump	P-101 main pump, P-602 cone bypass pump
M	Mixer / fitting	M-601 bypass mixing tee

Numbering: the hundreds digit denotes the equipment group (1 = tank, 2 = filtration, 3 = biological, 4 = disinfection, 5 = degassing, 6 = oxygenation, 7 = skimming, 8 = denitrification/temperature, 9 = advanced/utility); the tens+units denote the sequence within the group (01 = main, 02–09 = auxiliary/branch). Instrument prefixes: DO, pH, TT (temperature transmitter), LT (level transmitter), PDS (differential-pressure switch), AT (analysis transmitter), FT (flow transmitter).

Appendix D Species Parameter Table

iRAS includes nine species presets (including custom). The four most common are shown; the full set is in the software.

Species	Stage	Weight (g)	Mo	FCR	Prot %	Feed %	Dens	T°C	S‰	Depth
Salmon	Fry	0.1→50	6	1.0	50	3.0	30	12	0	1.5
	Juvenile	50→500	8	1.1	45	1.5	50	14	0	3.0

Species	Stage	Weight (g)	Mo	FCR	Prot %	Feed %	Dens	T°C	S‰	Depth
	Grow-out	500→2000	6	1.2	42	1.0	60	15	12	5.0
	Market	2000→5000	4	1.2	40	0.95	70	15	30	7.0
Turbot	Fry	3→50	4	0.9	55	3.0	25	16	28	0.8
	Grow-out	50→250	8	1.1	50	1.5	50	16	28	1.0
	Market	250→600	6	1.3	48	0.9	60	16	28	1.0
Bass	Fry	1→50	2	0.9	50	3.5	25	25	0	1.5
	Grow-out	50→250	4	1.1	45	1.8	40	25	0	2.5
	Market	250→600	4	1.3	40	1.2	50	25	0	3.0
Shrimp	Fry	0.01→1	1	1.0	45	10	2	30	15	1.5
	Early	1→10	1.5	1.3	42	5	8	30	15	2.0
	Late	10→30	1.5	1.6	38	3	15	30	15	2.5

Oxygen saturation target by species (%)

Species	Salmon	Turbot	Grouper	Bass	Mandarin	Eel	Tilapia	Shrimp
Target sat%	165	140	140	125	125	125	115	110

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Appendix F Disclaimer

The results of this tool (iRAS) and this manual are for engineering pre-estimation reference only.

- Real projects must combine on-site water-quality monitoring, batch trial data, local codes and professional engineering judgement.
- Biological parameters (FCR, protein content, etc.) vary significantly with species, feed and culture mode.
- Investment-estimate accuracy is about ±40%; actual cost depends on vendor quotations and competitive tendering.
- Operating cost excludes soft costs (labour, fingerlings, management, insurance); the all-in cost is typically 20–30% below whole-cost figures and these should be added separately.
- The two-node steady-state solver is literature-calibrated, but actual shop temperature/humidity depends on local airflow and layout and may differ from the model by 10–20%.
- The modular redundancy is based on the N+1 engineering convention; high-reliability projects may need N+2 or higher.
- The financial evaluation uses a single-point deterministic cash-flow model without inflation, exchange-rate risk or tax incentives; for complex financing or high-sensitivity projects, use specialised software (e.g. Monte-Carlo tools).
- The feasibility-report generator outputs structured content, not a finished report; design firms should paste it into their own template and add project identity, site conditions, detailed civil design and the specialist environmental/safety/energy assessments, which must be issued independently by qualified institutions.
- The system-discharge output is the settling-tank inlet concentration (sludge-laden, mass-conservation basis) and must not be compared directly with pipe-outlet effluent standards; settling-tank design and supernatant treatment are handled separately by the wastewater discipline.

- The authors and company accept no liability for engineering decisions made on the basis of this software's results.

— *End of Manual* —

iras.yis.no · github.com/chengsunmail/iras-en · polarlysinnovation@gmail.com